

Timing-Feedback Adaptive Scheduler for Split-6 Latency Reduction

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Abstract Section (before Introduction) Add a clear placeholder for an abstract:

Purpose: Briefly summarize the problem, method, results, and contribution.

Word count: typically 150—250 words.

Keywords: list 3—6 keywords.

I. Introduction

A. Introduction

Purpose: To explain what the paper is about, why the problem matters, and what the others authors have done to address it.

Typical content:

- Background and motivation
- Problem statement
- Goals of the paper
- Contributions (what's new or important)
- High-level overview of the approach or results

Example phrases:

- “In recent years, ...”
- “This paper addresses the problem of ...”
- “Our main contributions are ...”

B. Related Works

Purpose: To show how your work fits into the existing body of knowledge, and how it differs from or improves upon previous work.

Typical content:

- Summary of key prior research
- Strengths and limitations of existing approaches
- Comparison to your approach
- Justification for why a new approach is needed

Example phrases:

- “Several studies have explored ...”
- “Unlike [Author], we propose ...”
- “Prior work has not considered ...”

The main contributions of this paper are summarized as follows.

- Contribution 1.
- Contribution 2.
- Contribution 3.

The rest of the paper is organized as follows. Section II defines the system model/architecture considered in this paper.

The proposed analytical model/method is given in Section III. Section IV shows the numerical results. Concluding remarks are given in Section VI.

II. System Model

Figure 1 illustrates the system architecture of the proposed Timing-Feedback Adaptive Scheduler within a Split-6 RAN environment. The system comprises two main entities: the Virtual Network Function (VNF) and the Physical Network Function (PNF), connected via a packet-switched network.

The VNF hosts the upper layer protocol stack and the scheduler. To minimize transmission latency and ensure synchronization with the PNF, the VNF implements a feedback-based timing mechanism. The key components include:

- **Timing-Feedback Adaptive Module:** This module acts as the core controller. It receives timing feedback from the PNF, specifically the difference between the packet arrival time and the expected slot time (denoted as late by $N \mu s$ or early by $M \mu s$). It also incorporates internal scheduler timing metrics. Based on these inputs, it dynamically computes the optimal sleep duration for the VNF timing thread to align with the PNF's slot cycle.
- **VNF Timing Thread:** This thread is responsible for triggering the scheduler instances. It utilizes the corrected timing provided by the Adaptive Module to wake up at the precise moment, compensating for network jitter and processing delays.
- **Scheduler:** Detailed scheduling operations are performed here. Multiple scheduler instances generate the Transmission Time Interval (TTI) messages, which are then transmitted to the PNF.
- **PNF:** The Physical Network Function receives the TTI messages for radio transmission. It measures the arrival time of these messages and feeds this “Timing Info” back to the VNF, closing the control loop.

The primary objective of this system is to adaptively adjust the VNF's execution offset such that TTI messages arrive at the PNF within the valid reception window, thereby avoiding slot drops due to late arrival while minimizing buffering latency caused by excessive earliness.

III. Proposed Method

IV. Numerical/Simulation/Experimental Results

Computer simulations/Experiments were conducted to verify xxx (the effectiveness of the proposed analytical model).

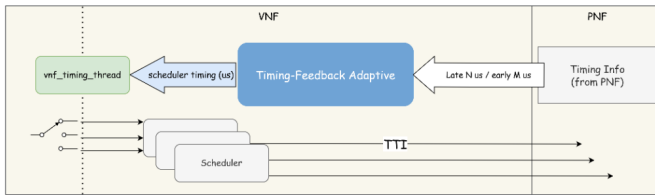


Fig. 1. System Model of the Timing-Feedback Adaptive Scheduler.

In the following figures, each point represented the average value of 10^5 samples. In each sample, we collect the statistics of xxx. Symbols and lines were used to show the simulation and analytical results, respectively.

XXX scenarios were investigated. Scenario I was designed to demonstrate/verify the accuracy/effectiveness of Contribution 1.

A. Contribution 1

Fig. 1.1: Purpose X axis: Y axis:

B. Contribution 2

Fig. 2.1: X axis: Y axis:

C. Contribution 3

Fig. 3.1: X axis: Y axis:

References

[1]